8 July 1975
Superseding
MIL-T-2487A
16 February 1959

MILITARY SPECIFICATION
TEST SET TS-27( )/TSM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Test Set, TS-27( )/TSM, a portable battery operated instrument designed to evaluate electrical characteristics of communication lines, such as conductor and insulation resistance, capacitance, and location of shorts, crosses, opens and grounds. The ZM-7( )/TSM also provides an alternating current bridge for measurement of capacitance and the location of opens.

2. APPLICABLE DOCUMENTS

2.1 Documents. The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, Wrapping
NN-P-71	Pallet, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple Wall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	Strapping, Nonmetallic, (and Connectors)
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant, (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

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MILITARY

MIL-E-75	Electron, Tube Packing, and Container Marking, General Specification for
MIL-P-116	Preservation-packaging, methods of
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finish for Ground Signal Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-781	Reliability Tests; Exponetial Distribution
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-DL-126023	Drawing List - Test Set TS-27( )/TSM
SC-A-46439	List of Accessories for Package Tester

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

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3. REQUIREMENTS

3.1 First Article Samples. When specified in the contract or purchase order, the contractor shall furnish First Article Units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with SC-DL-126023 and the requirements specified herein.

3.3 Parts, materials and processes-general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein shall apply.

3.4 Finish, protective. The equipment shall be finished in accordance with MIL-F-14072. This includes finish of hardware; such as screws, etc., and necessary touch-up after mounting. The final paint film on Type I surfaces shall be final Film E, semi-gloss light gray enamel, conforming to MIL-F-14072.

3.5 Marking. Marking shall conform to MIL-M-13231.

3.5.1 Visibility. Wherever practicable, parts and assemblies shall be so mounted that their identification markings shall be readily visible with a minimum disassembly of the equipment.

3.5.2 Serial numbers. Serial numbers are required.

3.6 Preconditioning.

3.6.1 Bounce preconditioning. Each equipment produced and first article shall be subjected to the bounce preconditioning of 4.6.1. After bounce preconditioning the equipment shall have no loose or broken parts and controls. If a failure occurs, the loose or broken part or control shall be repaired or replaced and the test repeated. Bounce preconditioning shall be performed prior to electrical preconditioning and Group A inspection.

3.6.2 Electrical preconditioning. Each equipment produced and first article shall be be subjected to the electrical preconditioning of 4.6.2. Electrical preconditioning shall be performed after bounce preconditioning and prior to Group A inspection. The equipment shall be monitored for requirements of 3.7.4 during period of test.

3.7 Electrical.

3.7.1 Accuracy. The equipment shall meet the accuracy requirements listed below when tested in accordance with 4.7.1.

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MEASUREMENTS	RANGE	ACCURACY
Resistance	0 to 50 Megohms	ohms: ±2% ohms x 100: ±5%
Capacitance	.075 to 3.0 microfarads	±5%
Oscillator Frequency		20 cps ±3 cps
Waveform		Sinusoidal (see 3.7.2)

3.7.1.1 Resistance. When tested in accordance with paragraphs 4.7.1.1 each test set shall measure resistance to within the accuracies stated above.

3.7.1.2 Capacitance. When tested in accordance with paragraphs 4.7.1.2 each test set shall measure capacitance to within the accuracies stated above.

3.7.1.3 Frequency. When tested in accordance with paragraph 4.7.1.3 the frequency of the oscillator shall be within the limits given in 3.7.1.

3.7.2 Waveform. The waveform shall be sinusoidal in shape and shall not display sharp corners, flat surfaces, or depressions at the peak at the curve. See paragraph 4.7.2.

3.7.3 Dielectric strength and insulation resistance. When tested in accordance with paragraph 4.7.3 there shall be no evidence of dielectric breakdown and the insulation resistance shall be greater than 500 megohms.

3.7.4 Operational. When specified, the equipment shall be subjected to the following operational requirements:

FUNCTION	MEASUREMENT	REQ. PAR.	TEST PAR.
Resistance	3000 ohms	3.7.1.1	4.7.1.1
Resistance	100 ohms	3.7.1.1	4.7.1.1
Resistance	1.0 megohms	3.7.1.1	4.7.1.1
Capacitance	1.0 mfd	3.7.1.2	4.7.1.2
Capacitance	2.0 mfd	3.7.1.2	4.7.1.2
Frequency		3.7.1.3	4.7.1.3

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3.8 Environmental. The Test Set TS-27( )/TSM shall be constructed so that no fixed part shall become loose, cracked, flaked, pitted or blistered, no movable part or control shall be shifted in setting, position or adjustment, and no permanent degradation shall be caused in performance below that specified herein when subjected to the following environmental conditions.

3.8.1 Rain. The test set shall show no evidence of water penetration and shall meet the electrical requirements of 3.7.4 before and after rain test, no operation during rain test, when tested in accordance with 4.8.1.

3.8.2 Vibration. The test set shall show no evidence of breakage or loosening of parts, and shall meet the electrical requirements of 3.7.4 before and after vibration test, when tested in accordance with 4.8.2.

3.8.3 Bounce, loose cargo. The test set shall show no evidence of physical damage, except minor surface abrasion and shall meet the electrical requirements of 3.7.4 before and after bounce test, when tested in accordance with 4.8.3.

3.8.4 Shock, bench handling. The test set shall show no evidence of physical damage, except minor surface abrasion and shall meet the electrical requirements of 3.7.4 at times specified, when tested in accordance with 4.8.4.

3.8.5 Altitude. The test set shall be subjected to the altitudes specified and shall meet the electrical requirements of 3.7.4 at times specified, when tested in accordance with 4.8.5.

3.8.6 High temperature. The test set shall be subjected to the high temperature extremes of 125°F operating and 160°F storage and shall meet the electrical requirements of 3.7.4 at times specified, when tested in accordance with 4.8.6.

3.8.7 Low temperature. The test set shall be subjected to the low temperature extremes of -4°F operating and -65°F storage and shall meet the electrical requirements of 3.7.4 at times specified, when tested in accordance with 4.8.7.

3.8.8 Humidity. The test set shall exhibit no physical damage of parts and material, such as corrosion, rust, blistering, swelling, deterioration, etc., or electrical degradation, and shall meet the electrical requirements of 3.7.4 at times specified, when tested in accordance with 4.8.8.

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3.8.9 Salt fog. The test set shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface, and shall meet the electrical requirements of 3.7.4 at times specified, when tested in accordance with 4.8.9.

3.8.10 Fungus. The test set shall show no evidence of viable fungus and shall meet the electrical requirements of 3.7.4 before and after fungus test, when tested in accordance with 4.8.10.

3.9 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall meet Requirement 7 of MIL-STD-454. Inspection shall be in accordance with 4.9.

3.10 Workmanship. The test set shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454, and with the applicable portions of MIL-P-11268 as follows:

a. Adhesives.
b. Cables, cords, and wires, electrical.
c. Wire (harnessing).
d. Lubricants.
e. Plastic material and parts.
f. Rubber.
g. Tropicalization of material and parts.
h. Wiring and cabling.
i. Connectors (electrical).
j. Screws, other threaded devices, and related parts.
k. Tightening of threaded parts.
l. Riveting.
m. Brazing, soldering, and welding.
n. Welding.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (does not include preparation for delivery) (see 4.3).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.12).

4.3 First article. Unless otherwise specified in the contract or purchase order the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish three (3) first article units of the completely assembled Test Set TS-27( )/TSM.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, preconditioning specified in 4.6, and the inspections specified for Group A, Group B, and Group C (see Tables I, II, and III, respectively). The inspection shall be performed in the following order: 4.4, 4.6, Group A and Group B for all units; and designated units for specific subgroups within Group C in accordance with 4.5.3.1. After completion of Group C environmental tests, conforming units shall be reinspected and shall pass all Group A inspection.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery; with visual and mechanical inspection in accordance with Table I:

Item	Where required
Parts, materials and processes-general	3.3
Finish	3.4
Marking	3.5
Workmanship	3.10

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4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government sampling inspection shall be made available by the contractor.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105 using the general inspection levels, and AQL's indicated in Table I.

4.5.1.1 Order of inspection within Group A. GROUP A inspection shall be performed in an order satisfactory to the Government.

4.5.1.2 Lot size. Lot sizes shall be determined in accordance with the monthly production schedule as follows:

Production rate	Size of lot
500 or more per month	One week's production
51 to 499 per month	Two week's production
50 or less per month	One month's production

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Table I - Group A Inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual & mechanical	)		1 dphu*	4 dphu
Construction	3.2)		for the	for the
Parts, materials and processes-general	3.3)		group	group
Finish, protective	3.4)	4.11		
Marking	3.5)			
Workmanship	3.10)			
Electrical	)			
Accuracy	3.7.1	4.7.1	1 dphu	**
Resistance	3.7.1.1	4.7.1.1	for the	
Capacitance	3.7.1.2	4.7.1.2	group	
Frequency	3.7.1.3	4.7.1.3		
Waveform	3.7.2	4.7.2		

* Deficiencies per hundred units.
** All electrical deficiencies are major.

4.5.2 Group B inspection. Group B inspection shall normally be performed in inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table II and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sampling plans, for the AQL's listed in Table II, shall be as follows:

AQL	Inspection Level
2.5 dphu	S4
4.0 dphu	S4

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

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Table II - Group B Inspection

Inspection	Req. Para.	Test Para.	AQL
Interchangeability	3.9	4.9	4 dphu*
Dielectric Strength and Insulation Resistance	3.7.3	4.7.3	2.5 dphu

* Defects per hundred units

4.5.3 Group C inspection. This inspection shall consist of the inspection specified in Table III and shall be performed on sample units that have been subjected to and met all Group A and B tests. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection.

4.5.3.1.1 Three (3) samples of the complete equipment shall be selected at random, one (1) sample for Subgroup 1, one (1) sample for Subgroup 2 and 3, and one (1) sample for Subgroup 4 in Table III. The samples selected shall be at the start of the contract from the first quality conformance inspection lot. These samples shall constitute Group C requirements for first 100 units produced.

4.5.3.1.2 One (1) sample of the complete equipment shall be selected at random for Subgroup 5 in Table III. The sample selected shall be at the start of the contract from the first quality conformance inspection lot only. Sample not required for first article.

4.5.3.1.3 Thereafter, two (2) samples of the complete equipment shall be selected at random, one (1) sample for each Subgroup 1 and 2 in Table III from every 500 units and fraction thereof produced.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order shown in each Subgroup in Table III. Subgroup 3 inspection shall be performed on the sample after subjection to Subgroup 2.

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Table III - Group C Inspection

Inspection	Req Para.	Test Para.
Subgroup 1		
Rain	3.8.1	4.8.1
Vibration	3.8.2	4.8.2
Shock, Bench handling	3.8.4	4.8.4
Bounce, loose cargo	3.8.3	4.8.3
Subgroup 2		
Altitude	3.8.5	4.8.5
High temperature	3.8.6	4.8.6
Low temperature	3.8.7	4.8.7
Humidity	3.8.8	4.8.8
Subgroup 3		
Salt fog	3.8.9	4.8.9
Subgroup 4		
Fungus	3.8.10	4.8.10
Subgroup 5		
Construction	3.2	4.10

4.5.3.3 Evaluation of samples submitted to individual Group C inspection. After samples referenced in 4.5.3.1 have been subjected to an individual Group C inspection test and prior to subjection to the next Group C test, unless otherwise specified in the individual tests, the sample shall meet requirements of 3.7.4.

4.5.3.4 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group B and C sample Units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or purchase order provided all damage is repaired and the sample units are subjected to and pass Group A inspection.

4.5.5 Disposition of Group C samples. Unless otherwise specified, samples which have been subjected to and passed Group C inspection may be delivered on the contract or purchase order provided samples meet requirements of 4.5.4.

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4.6 Preconditioning.

4.6.1 Bounce preconditioning. Each test set produced and first article shall be placed in its normal operating position on the table of a package tester of suitable capacity as made by the L.A.B. Corporation, Skaneateles, New York, with accessories as per Model 1000-SC and SC-A-46439, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the equipment under test to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute and shall meet the requirements of 3.6.1.

4.6.2 Electrical preconditioning. Each equipment produced and first article shall be placed in its normal operating position on a test table or in a test rack with room temperature at 77° ±5°F, with specified operating voltage applied from a suitable power source. With all controls in the normal operating position, the equipment shall be turned on and remain on for a continuous 24 hour period and monitored. An external dc power supply may be used in lieu of batteries during test. In case of failure, the equipment shall be repaired, recalibrated, and the electrical preconditioning test repeated for the failed equipment. If required in the contract or purchase order, a failed item analysis report shall be submitted on each failure with contents of report in accordance with 5.10.5 of MIL-STD-781. Each equipment shall meet the requirements of 3.6.2.

4.7 Electrical tests. Unless otherwise specified, test equipment shall be calibrated to an accuracy of at least five times the accuracy of the item under test.

4.7.1 Accuracy test.

4.7.1.1 Resistance.

4.7.1.1.1 Resistance measurements with "PR-1" and "PR-2" circuits. Connect resistors, of the values and accuracies specified in the first column of the following table, successively between binding posts "PR-1". Throw key "PR-1" to the "Test" position and set selector switch on the "T-R" position. Then depress the corresponding key, (indicated in the second column of the table.); galvanometer balance shall be obtained within the limits specified in the third column of the table, on "Ohms" and "megohms" scales, as indicated. The procedure shall be repeated with the resistors connected successively between binding posts "PR-2", with Key "PR-2" thrown to the "Test" position and with selector switch set on the "T-R" position.

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Table IV

Resistor Value, Ohms	Depress Key	Reading "Ohms" Scale
10000 ±1%	"Ohms x 100"	100 ±2.0
3000 ±1%	"Ohms x 100"	30 ±0.6
30000 ±1%	"Ohms x 100"	300 ±6.0
30 ±1%	"Ohms"	30 ±0.6
100 ±1%	"Ohms"	100 ±2.0
300 ±1%	"Ohms"	300 ±6.0
Resistor Value, Megohms	**Depress Key**	**Reading "Megohms" Scale**
0.3 ±1%	"Megohms"	0.3 ±0.015
1.0 ±1%	"Megohms"	1.0 ±0.050
3.0 ±1%	"Megohms"	3.0 ±0.15

4.7.1.1.2 Ground Location with "PR-1" and "PR-2" circuits.

4.7.1.1.2.1 Ground location with "PR-1" circuit. Connect resistors of 300 ohms and 100 ohms, (each value plus or minus one per cent) in series across binding posts "PR-1". Connect junction point of two resistors to binding post "G". Throw key "PR-1" to "Test" position, set selector switch at "Gnds" position and depress "Ohms" key. The galvanometer shall balance on the "O" position when the slide wire dial is adjusted to read 50, plus or minus 0.5 division, on the "Grounds" scale. It should be noted whether the balance is obtained with a dial setting on the right or left of the mid-point of the scale. The procedure shall be repeated with the connections of the series resistors to binding posts "PR-1" interchanged. The galvanometer shall balance on the "O" position when the slide wire dial is adjusted to read 50, plus or minus 0.5 division, on the "Grounds" scale. It shall be ascertained that the balance is obtained with a dial setting on the opposite half of the scale from that obtained above.

4.7.1.1.2.2 Ground location with "PR-2" circuit. Repeat the procedure of 4.7.1.1.2.1 with the series resistor combination connected across binding posts "PR-2" and with key "PR-2" thrown to "Test" position. Galvanometer balance shall be obtained with a dial setting, on either half of the "Grounds" scale of 50 plus or minus 0.5 division. This procedure

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shall be repeated with the connections of the series resistors to binding posts "PR-2" interchanged. Galvanometer balance shall be obtained with a dial setting, on the "Grounds" scale, of 50 plus or minus 0.5 division. It shall be ascertained that the balance is obtained with a dial setting on the opposite half of the scale from that obtained above.

4.7.1.2 Capacitance.

4.7.1.2.1 With no external circuits connected to the bridge, keys "PR-1" and "PR-2" in their "NORMAL" positions, and the slide wire dial is set at "O" on "Opens" scale, the galvanometer shall give no deflection when "Opens" key is depressed. Then test the following combination of capacitors and resistors (consecutively and in the order given) as specified in 4.7.1.2.1 and 4.7.1.2.2, between binding posts "PR-1" with key "PR-1" in "Test" position and with selector switch in "T-R" position. A galvanometer balance shall be obtained when the slide wire dial is set at a point that is within the limits specified. Each capacitance value shall be within the limits of plus or minus one percent.

Capacitance	Balance point on "Opens" scale
0.075 mfd.	12.5 to 13.5
0.20 mfd.	27.5 to 29.5
0.50 mfd.	49.0 to 51.0
1.00 mfd.	65.5 to 67.5
2.00 mfd.	79.0 to 81.0

4.7.1.2.2 The test of paragraph 4.7.1.2.1 shall be repeated with resistances, within ±5 per cent of the values specified below, connected in parallel with the several capacitances. A galvanometer balance shall be obtained, for each of these combinations, at the same slide-wire dial setting as that which was obtained for the corresponding point in the test of paragraph 4.7.1.2.1.

Capacitance	Paralleling resistance
0.075 mfd.	400,000 ohms
0.20 mfd.	150,000 ohms
0.50 mfd.	60,000 ohms
1.00 mfd.	50,000 ohms
2.00 mfd.	15,000 ohms

4.7.1.3 Frequency. With binding posts T and R of "PR-1" or "PR-2" connected to a low frequency counter, Switch "PR-1" or "PR-2" respectively in "Test" position and "Selector" switch in "T-R" position, requirements of 3.7.1.3 shall be met.

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4.7.2 Waveform. With connections and switch settings identical to 4.7.1.3 except that the counter is replaced by an ordinary laboratory oscilloscope, the requirements of 3.7.2 shall be met.

4.7.3 Dielectric strength and insulation resistance. Unless otherwise specified the dielectric strength of all circuits shall be tested with a potential of 400 volts d.c., or peak 60 cycle a.c. (Test shall be performed with Tubes, galvanometer and batteries removed and batteries replaced by short circuit connectors.) The potential shall be applied between binding posts "PR-1" and "PR-2" wired together; and binding post "G" electrically connected to metal frame of chassis. With selector switch set on "T-R" and keys "PR-1" and "PR-2" in the "Test" position, the test potential shall be applied for a period of 10 seconds. Immediately after the conclusion of the above test, and using the same test points, the insulation resistance reading shall not be less than 500 megohms. (see 3.7.3)

4.8 Environmental. The test set shall be tested in accordance with the following environmental conditions.

4.8.1 Rain. The test set, in its normal operating configuration, shall be subjected to the rain test specified in Procedure I, Method 506 of MIL-STD-810. Inability of the equipment to meet the requirements of 3.8.1 shall constitute failure of this test.

4.8.2 Vibration. The test set shall be subjected to the vibration test specified in Part 1, Procedure IX, Method 514 of MIL-STD-810 in the horizontal, parallel to the major horizontal axis of the equipment; horizontal, at right angles to the major horizontal axis of the equipment; and vertically to the major axis of the equipment. Inability of the equipment to meet the requirements of 3.8.2 shall constitute failure of this test.

4.8.3 Bounce, loose cargo. The test set, as prepared for field transportation, shall be subjected to the loose cargo bounce test specified in Part 2, Procedure IX, Method 514 of MIL-STD-810. Inability of the equipment to meet the requirements of 3.8.3 shall constitute failure of this test.

4.8.4 Shock, bench handling. The test set shall be subjected to the bench handling shock test specified in Procedure V, Method 516 of MIL-STD-810. Inability of the equipment to meet the requirements of 3.8.4 shall constitute failure of this test.

4.8.5 Altitude. The test set, in its normal operating configuration shall be subjected to the altitude test specified in Procedure I, Method 500 of MIL-STD-810, with altitude in Step 3 held for 1 hour. Inability of the equipment to meet the requirements of 3.8.5 shall constitute failure of this test.

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4.8.6 High temperature. The test set, in its normal operating configuration, shall be subjected to the high temperature test specified in Procedure I, Method 501 of MIL-STD-810, with temperature in Step 4 held for 12 hours. Inability of the equipment to meet the requirements of 3.8.6 shall constitute failure of this test.

4.8.7 Low temperature. The test set, in its normal operating configuration shall be subjected to the low temperature test specified in Procedure I, Method 502 of MIL-STD-810, with temperature in Step 2 held for 48 hours and in Step 4, held for 12 hours. Inability of the equipment to meet the requirements of 3.8.7 shall constitute failure of this test.

4.8.8 Humidity. The test set, in its normal operating configuration, shall be subjected to the humidity test specified in Procedure II, Method 507 of MIL-STD-810. Inability of the equipment to meet the requirements of 3.8.8 shall constitutes failure of this test.

4.8.9 Salt fog. The test set, in its normal operating configuration, shall be exposed to the salt fog test specified in Procedure I, Method 509 of MIL-STD-810 for 48 hours. Inability of the equipment to meet the requirements of 3.8.9 shall constitute failure of this test.

4.8.10 Fungus. The equipment shall be subjected to the fungus test specified in Method 508, Procedure I, of MIL-STD-810. There shall be abundant growth colonization (6.6) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the equipment to meet the requirements of 3.8.10 shall constitute failure of this test.

4.9 Interchangeability. The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirement of 3.9. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as case, covers, and mountings.

d. Location and spacing of test terminals.

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4.10 Construction. The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical examination of the equipment on contract against the drawings cited for construction in 3.2. The EVR shall consist of the following:

4.10.1 An examination to establish that as built, equipment (unit, assembly, module or part) is in accordance with the cited end product drawings.

4.10.2 An examination to establish that as built, configuration of a unit, assembly, module or part meets the acceptance requirements (other than equipment specification performance) specified for each unit, assembly, module or part. The examination shall record all acceptance test methods used together with resulting verification test data.

4.10.3 The contractor shall compile an EVR report containing the following information:

a. Identification of unit, assembly, module or part and details shall be in accordance with 4.10.1 and 4.10.2.

b. Discrepancies noted.

c. Corrective action taken.

4.10.4 The contractor shall furnish the Government Quality Assurance Representative a copy of the EVR report no later than fifteen (15) days prior to the submission of the first production unit or lot. The results of the EVR shall be subject to inspection by Government personnel at the time the first unit or production lot is offered for acceptance. Government inspection will be to the depth and extent necessary to demonstrate that the as-built hardware is in accordance with the cited drawings. The contractor shall provide the following:

a. Segregation of units, assemblies, modules and parts to permit Government personnel access for detailed inspection.

b. As required, responsible personnel from each functional department for necessary discussions in their respective areas.

c. Adequate administrative support for the EVR.

4.11 Visual and mechanical. The equipment shall be subjected to a thorough visual and mechanical inspection. The defects shall be classified in accordance with MIL-STD-252, and the documents referenced in requirement paragraphs listed in Table I, Group A inspection.

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4.12 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1.1 Cleaning. Test Set TS-27( )/TSM shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Test Set TS-27( )/TSM shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Electron tubes (spares). Spare tubes shall be packaged individually in accordance with MIL-E-75.

5.1.1.4.3 Chart sets. Chart sets shall be placed in the compartment in the lid and secured.

5.1.1.4.4 Test leads. Each test lead shall be packaged method III. Coil each lead into a coil of proportionate dimensions and tie in three places with cotton twine conforming to T-T-871.

5.1.1.4.5 Test Set TS-27( )/TSM. Each test set shall be packaged method III. Place the spare parts, packaged as specified in 5.1.1.4.2 through 5.1.1.4.4, within the compartment provided in the case cover. Close and secure all fastenings. Cushion the case on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned case within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

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5.1.2 Level C. Test set TS-27( )/TSM. Each clean and dry test set shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Palletized load. A quantity of test sets, packaged as specified in 5.1, shall be placed on pallets in conformance with MIL-STD-147, load type I, with a weather-resistant fiberboard cap (storage aid 4) positioned over the load.

5.2.1.2 Less than palletized load. When quantities per destination are less than a pallet load a quantity of test sets, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2. Closure shall be in accordance with the appendix of the box specification.

5.2.2 Level B.

5.2.2.1 Palletized load. A quantity of test sets, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1, except that the fiberboard caps shall be class domestic.

5.2.2.2 Less than palletized load. When quantities per destination are less than a pallet load, a quantity of test sets, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 1. Closure shall be in accordance with the appendix of the box specification.

5.2.3 Level C.

5.2.3.1 Palletized load. A quantity of test sets, packaged as specified in 5.1, shall be palletized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulation.

5.2.3.2 Less than palletized load. Test sets, packaged as specified in 5.1, shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package, interior container and palletized load shall be marked in accordance with MIL-STD-129.

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5.4 General.

5.4.1 Army procurements (see 5.2.1, 5.2.2 and 5.2.3).

5.4.1.1 Level A packing shall be as follows:

5.4.1.1.1 Palletized load. A quantity of test sets, packaged as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film.

5.4.1.1.2 Less than palletized load. When quantities per destination are less than a pallet load, the item packaged as specified in 5.1, shall be waterproofed with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification, or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.4.1.2 Level B packing shall be as follows:

5.4.1.2.1 Palletized load. A quantity of test sets, packaged as specified in 5.1, shall be palletized as specified in 5.4.1.1.1.

5.4.1.2.2 Less than palletized load. When quantities per destination are less than a pallet load, a quantity of test sets, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

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5.4.1.3 Level C packing shall be as follows:

5.4.1.3.1 Palletized load. A quantity of test sets, packaged as specified in 5.1, shall be palletized as specified in 5.4.1.1.1, except that the fiberboard caps shall be class domestic.

5.4.1.3.2 Less than palletized load. When quantities per destination are less than a pallet load, a quantity of test sets, packaged as specified in 5.1, shall be packed as specified in 5.4.1.2.2, except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.4.2 Navy procurements. For Navy procurements the use of polystyrene loose fill material (such as strips, strands and beads) is prohibited for packaging and packing applications.

6. NOTES

6.1 Intended use. Test Set TS-27( )/TSM is intended for use at operator/crew level.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level of preservation, packaging, packing and marking (see Section 5).

c. If rough handling and functional tests are required.

d. Quantity of first articles required.

e. Place of final inspection.

f. Technical literature required.

g. Quantity of tools and running spare parts required.

6.3 Nomenclature and type designation. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. The parentheses in the type designation will be deleted or replaced by a letter identifying the particular design for example: SG-12( )/U to SG-12F/U.

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6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.4).

6.5 Sampling inspection. Sampling by the Government will be limited to the amount deemed necessary to determine compliance with the contract or purchase order and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of sampling inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.6 Definitions.

6.6.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.6.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.6.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.6.4 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.6.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

Custodian:
Army - EL

Preparing activity:
Army - EL

User Activity:
Army - AT, MD, SC

Project No. 6625-0510

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-2487B TEST SET TS-27( )/TSM (EL)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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